

David Gusmao

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EDUCATION

University of Central Florida

B.S. in Computer Science, GPA: 3.38 | Passed the CS Foundation Exam

Expected December 2026

Orlando, FL

EXPERIENCE

Project Manager & Technical Lead: Knights Counsel

Jun 2026 – Present

UCF Senior Design, Sponsored Capstone

Orlando, FL

- Lead a team building a permission-aware AI document-retrieval assistant for a university legal office, deployed in the client's Microsoft 365 tenant against a live document corpus.
- Designed and built the retrieval agent end to end: two-agent topology, specified behavioral requirements, grounding and citation enforcement, and a version-controlled configuration baseline.
- Built an 18-case evaluation suite and a custom grading rubric after finding the platform's built-in grader scored answer rate rather than designed behavior; results redirected the team's critical path.
- Co-authored the project's technical white paper; sole author of the system design and behavior specification sections.

Undergraduate Research Assistant: Hardening Build Systems

Oct 2025 – May 2026

University of Central Florida

Orlando, FL

- Diagnosed build failures by hand across six build systems (Autotools, Make, CMake, Maven, Gradle, Ant), including undeclared JDK version and compiler-flag requirements.
- Built the tooling that turns raw experimental output into LaTeX-ready tables and pgfplots figures used in project reporting.
- Supported reproducible workflows by organizing datasets, validating results, and documenting analysis steps.

Research Assistant: Human Factors Group Study

Feb 2025 – May 2026

University of Central Florida

Orlando, FL

- Ran experimental sessions studying how programmers debug code in groups, contributing to data collection workflow and study setup.
- Instrumented participants with biometric equipment (heart rate, respiration, skin conductance, gaze tracking, prefrontal cortex blood oxygenation) to improve measurement quality across sessions.

PROJECTS

Build-Recipe Inference Agent | *Python, Docker, Anthropic API, Linux*

May 2026 – Present

- Building a containerized LLM agent that infers and executes build recipes for unfamiliar source archives, selecting the build system, installing system and toolchain dependencies, and running each build phase in a fresh isolated container.
- Designing a cross-tool comparison running three build-inference systems over three program corpora to separate genuine capability from benchmark fit, evaluated against a hand-verified corpus of ~370 build recipes.

PocketProfessors | *React, TypeScript, Node.js, Express, MongoDB, JWT*

Jul 2025

- Led frontend UX for pack-opening and collection views: responsive grid, focus states, and ARIA labeling; collaborated on API shape and the JWT refresh lifecycle. Deployed at pocketprofessors.com.

FitnessFunctions | *PHP, MySQL, HTML/CSS, JavaScript*

Nov 2025

- Solo-built role-based class enrollment system with admin and member dashboards, instructor and class CRUD, and capacity-aware enrollment backed by MySQL.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, PHP, C++, HTML, CSS, Bash, LaTeX

Front-End: React, Node.js, Express, Fluent UI, semantic HTML, responsive CSS (Flexbox/Grid), WCAG accessibility patterns

AI & Agent Engineering: LLM agent design, retrieval-augmented generation, grounding and citation enforcement, guardrail and refusal design, tool-use and function-calling interfaces, structured JSON output contracts, Anthropic API, OpenAI API, Microsoft Copilot Studio, Microsoft Power Platform (Power Automate)

Evaluation & Research: evaluation suite and rubric design, regression testing for non-deterministic systems, run-to-run variance analysis, pass@k, precision/recall, ground-truth construction, failure taxonomy, experimental design, literature synthesis

Systems & Infrastructure: Docker, Linux/WSL2, MongoDB, MySQL, Apache (LAMP), MERN, build systems (Autotools, Make, CMake, Maven, Gradle, Ant), SharePoint Online, Microsoft Entra ID, architecture decision records, technical writing

Tools & Practice: Git/GitHub, Jira, Confluence, Postman, VS Code, Overleaf, architecture decision records, technical writing

AWARDS & CERTIFICATIONS

Awards: 3rd Place, UCF Horse Plinko Cyber Defense Competition (Oct 2024); Bright Futures Academic Scholarship

Certifications: Human Subjects Research: Social/Behavioral (Group 2); Responsible Conduct of Research (Engineers); Certified Internet Business Associate